

Problem Set 5 for Fourier transforms. Due Friday October 24 at 11:59 PM

Problem 0: Please hand-write the numbers 0 through 9, scan, and compress each down to a 28x28 array of pixel brightnesses that go from 0 to 255. Put into a single array of size (10,28,28) and save as 'digits_[your student id]_week5.npy'. You can get a half bonus point if you write out 5 sets and save as a (50,28,28) array with the numbers ordered (0..9,0..9,0..9), and a full bonus point if you write out ten sets.

Problem 1: Write a function that will shift an array by an arbitrary amount using an FFT-based convolution (yes, I know there are easier ways to do this). The function should take 2 arguments - an array, and an amount by which to shift the array. Plot a gaussian that started in the centre of the array shifted by half the array length.

Problem 2: a) The correlation function $h(y) = f \star g$ is $h(y) = \int f(x)g(x+y)dx$. Through a proof similar to the one for the convolution theorem, one can show $f \star g = \text{ift}(\text{dft}(f) * \text{conj}(\text{dft}(g)))$. Write a routine to take the correlation function of two arrays. Plot the correlation function of a Gaussian with itself.

b) using these results, write a routine to take the correlation function of a Gaussian (shifted by an arbitrary amount) with itself. How does the correlation function depend on the shift? Does this surprise you?

Problem 3: The circulant (wrap-around) nature of the dft can sometimes be problematic. Write a routine to take an FFT-based convolution of two arrays without any danger of wrapping around. You may wish to add zeros to the end of the input arrays.

Problem 4: DFTs work very nicely out of the box when there are an integer number of periods of a wave in the region analyzed. Sadly, when we are dealing with real data, we usually are forced to analyze a finite chunk of data, and there will in general be no particular relation between the frequencies in the data and the interval we're analyzing. We'll look at the effects of this a bit now.

a) Show that:

$$\sum_{x=0}^{N-1} \exp(-2\pi i k x / N) = \frac{1 - \exp(-2\pi i k)}{1 - \exp(-2\pi i k / N)}$$

It may help to recognize that the sum can be re-written $\sum \alpha^x$ where $\alpha = \exp(-2\pi i k / N)$ so we can treat it as the sum of a geometric series.

b) Show that this approaches N as k approaches zero, and is zero for any integer k that is not a multiple of N .

c) We can use this to analytically write down the DFT of a non-integer sine wave/complex exponential (note that

$$\sum \exp(2\pi i k x / N) \exp(-2\pi i k' x / N) = \sum \exp(2\pi i x (k - k') / N)$$

. Pick a non-integer value of k and plot your analytic estimate of the DFT. Show that the FFT agrees (to within machine precision) with your analytic estimate. Normally, we think of the Fourier transform of a pure sine wave to be a delta function. Are we close to that? This phenomenon is usually known as *spectral leakage*. Note that the leakage is set by $k - k'$, so if I shift my frequency by an integer number of periods, the leakage pattern remains the same, just centered on the new frequency.

d) A common tool to get around this is the use of *window functions*. The leakage essentially comes from the fact that we have a sharp jump at the edge of the interval. If we multiply our input data by a function that goes to zero at the edges, this cancels out the jump, and so prevents the leakage from the jumps at the edges. Of course, since we have multiplied by the window in real space, we have convolved by it in Fourier space. One simple window we will use is $0.5 - 0.5 \cos(2\pi x / N)$ (there are many, many choices - over 30 on wikipedia). Show that when we multiply by this window, the spectral leakage for a non-integer period sine wave drops dramatically.

e) Show that the Fourier transform of the window is $[N/2 \ N/4 \ 0 \ \dots \ 0 \ N/4]$ (either numerically or analytically). Use this to show that you can get the windowed Fourier transform by appropriate combinations of each point in the unwindowed Fourier transform and its immediate neighbors (you may need to be careful with signs here, since if you work through the math, some of the transforms need to be inverse FFTs). The choice of suitable windows is as much art as science (and depends on the details of what you're most concerned about), but I hope this gives a flavor of what's going on and will be useful for matched filter results.